UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Faculty of Engineering and IT

Assignment Cover Sheet

Subject: 41277 Control Design

Assignment Number: Demonstration Practical 2 **Date Submitted:** [TODO: dd/mm/2026]

Assignment Title: State-Feedback Control

Student Name(s) and Number(s) **Tutorial Group:** 02

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Statement of Collaboration:

[TODO: Describe how the team collaborated on Q1–Q3.]

Demonstration Practice 2

Student Number	Student Name	Overall Contribution Weighting From 0 to 100%
24767939	Jaylen Avtarovski	100%
24817532	Rhyse Williams	100%
24949885	Elina Kim	100%

Each member's individual mark is the group mark scaled by their contribution level. If not indicated, all students get the same mark.

Answers to the Assignment Questions:

Q1: State-Space Model (40 marks)

Q1.1) System Modelling

Based on [1], the QUBE-Servo 2 is modelled as

$$\dot{x}(t) = A_c\, x(t) + B_c\, u(t), \tag{1a}$$
$$y(t) = C_c\, x(t) + D_c\, u(t), \tag{1b}$$

with $n =$ **[TODO:**n**]** states, $m =$ **[TODO:**m**]** inputs and $p =$ **[TODO:**p**]** outputs.

[TODO: Derivation of the motor/load dynamics and the state vector $x(t)$.**]**

Q1.2) System Parameters

The physical parameters of the QUBE-Servo 2 are given in Table 1, taken from the manufacturer's user manual [3].

Table 1: QUBE-Servo 2 system parameters [3].

Symbol	Description	Value
DC motor		
R_m	Terminal resistance	$8.4\,\Omega$
k_t	Torque constant	$0.042\,\mathrm{N\,m/A}$
k_m	Motor back-emf constant	$0.042\,\mathrm{V/(rad/s)}$
J_m	Rotor inertia	$4.0 \times 10^{-6}\,\mathrm{kg\,m^2}$
L_m	Rotor inductance	$1.16\,\mathrm{mH}$
m_h	Module attachment hub mass	$0.0106\,\mathrm{kg}$
r_h	Module attachment hub radius	$0.0111\,\mathrm{m}$
J_h	Module attachment hub inertia	$0.6 \times 10^{-6}\,\mathrm{kg\,m^2}$
Inertia disc module		
m_d	Disc mass	$0.053\,\mathrm{kg}$
r_d	Disc radius	$0.0248\,\mathrm{m}$
Rotary pendulum module		
m_r	Rotary arm mass	$0.095\,\mathrm{kg}$
L_r	Rotary arm length	$0.085\,\mathrm{m}$
m_p	Pendulum link mass	$0.024\,\mathrm{kg}$
L_p	Pendulum link length	$0.129\,\mathrm{m}$
Encoders and amplifier		
	Encoder resolution (quadrature)	$0.00307\,\mathrm{rad/count}$
	Recommended output voltage	$\pm 10\,\mathrm{V}$

[TODO: Keep only the rows for the module used; add any identified parameters (e.g. viscous friction) and the identification method.**]**

Q1.3) System Model Discretisation

A sampling time of T_s = **[TODO:**T_s**]** is selected because **[TODO:** justification, e.g. relative to the closed-loop bandwidth**]**. Using a zero-order hold (`c2d`), the discrete-time model is

$$x(k+1) = A\,x(k) + B\,u(k), \tag{2a}$$
$$y(k) = C\,x(k) + D\,u(k). \tag{2b}$$

[TODO: Numerical A, B, C, D.**]**

Q2: State-Feedback Control (SFC) **(40 marks)**

Q2.1) Controllability

For this question, the controllability matrix of the continuous-time model is

$$\mathcal{C} = \begin{bmatrix} B_c & A_cB_c & \cdots & A_c^{n-1}B_c \end{bmatrix}. \tag{3}$$

[TODO: Numerical $\mathcal{C}$ (`ctrb`), its rank and justification.**]**

Q2.2) State and Input References (Equilibrium Point)

Given the desired output reference $y^\star$, the steady-state references $x^\star$ and $u^\star$ satisfy

$$\begin{bmatrix} A - I & B \\ C & D \end{bmatrix} \begin{bmatrix} x^\star \\ u^\star \end{bmatrix} = \begin{bmatrix} 0 \\ y^\star \end{bmatrix}. \tag{4}$$

[TODO: Solve for $x^\star$, $u^\star$ for your setup.**]**

Q2.3) Closed-Loop Eigenvalues

The control requirements are an overshoot below 15 % and a settling time below 5 s. The damping ratio and natural frequency follow from

$$\zeta = \frac{-\ln(OS/100)}{\sqrt{\pi^2 + \ln^2(OS/100)}}, \qquad \omega_n = \frac{4}{\zeta\, t_s}, \tag{5}$$

and the discrete-time eigenvalues from $z_i = e^{s_i T_s}$.

[TODO: Chosen s_i and z_i for all n eigenvalues, with margin.**]**

Q2.4) State-Feedback Gain

For the control law

$$u(k) = -F\big(x(k) - x^\star\big) + u^\star, \tag{6}$$

[TODO: F from `place(A,B,pz)` and verification of eig$(A - BF)$.**]**

Q2.5) Closed-Loop Implementation

[Insert `assets/images/fig_q25_linear.png` here]

Figure 1: Closed-loop response of the linear model under state feedback.

[TODO: Comment on Figure 1: overshoot, settling time, steady-state error and control effort versus requirements. Add zooms.**]**

Q3: Luenberger Observer Design **(20 marks)**

Q3.1) Observability

The observability matrix of the continuous-time model is

$$\mathcal{O} = \begin{bmatrix} C_c \\ C_c A_c \\ \vdots \\ C_c A_c^{n-1} \end{bmatrix}. \tag{7}$$

[TODO: Numerical $\mathcal{O}$ (`obsv`), its rank and justification.**]**

Q3.2) Observer Eigenvalues

[TODO: Observer eigenvalues (typically several times faster than the SFC eigenvalues) in continuous and discrete time.**]**

Q3.3) Observer Gain

For the estimator

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + L\big(y(k) - \hat{y}(k)\big), \tag{8}$$

[TODO: L from `place(A',C',po_z)'` and verification of eig$(A - LC)$.**]**

Q3.4) Closed-Loop Implementation

The controller now uses the estimated state:

$$u(k) = -F\big(\hat{x}(k) - x^\star\big) + u^\star. \tag{9}$$

[Insert `assets/images/fig_q34_estimation.png` *here]*

Figure 2: True and estimated states with $\hat{x}(0) \neq x(0)$.

[Insert `assets/images/fig_q34_closed_loop.png` *here]*

Figure 3: Closed-loop response with observer-based state feedback.

[TODO: Comment on Figures 2 and 3: observer convergence and closed-loop performance against requirements.**]**

References

[1] R. P. Aguilera, *Lecture Notes on State-Feedback Control*, University of Technology Sydney, Australia, 2026.

[2] B. Messner and D. Tilbury, *Control Tutorials for MATLAB and Simulink*, University of Michigan, USA. [Online]. Available: http://ctms.engin.umich.edu/CTMS

[3] Quanser Inc., *QUBE-Servo 2 User Manual*, v1.0, Markham, ON, Canada.